

AMIR HOSEIN HEYDAR

Systematic Trading Research | Strategy Robustness & Portfolio Risk

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PROFILE

Independent systematic trading researcher with approximately five years of self-directed market research. Focuses on strategy evaluation, robustness, risk architecture, and capital-allocation decisions under uncertainty. Uses quantitative tooling to turn market hypotheses into testable systems; strongest discretionary market experience is intraday gold, with broader research across equities, FX, and commodities.

CORE INVESTMENT & RESEARCH CAPABILITIES

- **Strategy evaluation:** starts with why an edge should exist, then challenges hidden assumptions, regime dependence, capacity, costs, and portfolio interaction.
- **Risk architecture:** evaluates drawdown, volatility, correlation, concentration, liquidity, tail risk, leverage, execution, model risk, and operational fragility.
- **Capital allocation:** applies dynamic exposure, volatility targeting, capital scaling, and context-aware concentration/diversification rather than diversification as an end in itself.
- **Validation:** out-of-sample, walk-forward, Monte Carlo, parameter sensitivity, stress testing, cross-market validation, and regime testing.
- **Hybrid decision systems:** human judgment for context, research questions, and strategy selection; systematic processes for repeatable execution, sizing, monitoring, and risk control.

EXPERIENCE

Independent Systematic Trading Researcher | Self-Directed

2021 - Present

- Built an end-to-end research process spanning market observation, hypothesis formation, strategy logic, quantitative validation, risk design, execution assumptions, and post-failure review.
- Used a research universe of **10,000+ strategy/hypothesis variations** primarily as a falsification and comparison space - not a search for the best historical equity curve - across trend, mean-reversion, breakout, volatility, and regime-adaptive families.
- Developed a failure-first process: when a strategy breaks across a market, timeframe, or regime, investigates which assumption failed before adding filters, parameters, or complexity.
- Evaluates capital suitability through drawdown/recovery, payoff asymmetry, tail behavior, capacity, turnover, costs, correlation, and regime stability rather than win rate alone.
- Personal-capital and proprietary-trading experience; maintains a developing live/forward record while keeping realized results clearly separated from research backtests.
- Led a five-person marketing team outside trading, with hands-on experience in team direction, systems building, and execution accountability.

SELECTED RESEARCH FINDINGS

- **Trend-following robustness:** in my research universe, trend/continuation ideas remained more coherent than many mean-reversion and counter-trend variants as market, timeframe, regime, and parameter assumptions changed.
- **Prediction complexity:** moved increasingly toward confirmation/participation frameworks after repeated cases where added predictive complexity improved historical fit but increased hidden fragility and research-complexity cost.
- **Failure as data:** treats cross-market, cross-timeframe, and regime failure as evidence about hidden assumptions; the first response is diagnosis and possible rejection, not automatic re-optimization.

SELECTED PROJECTS

Decision Alpha Lab | Public quantitative research platform

github.com/bsAmirHeydar/decision-alpha-lab

- Public hypothesis-driven research platform for strategy validation, regime analysis, risk/execution governance, and explicit separation of research evidence from capital authority.
- Published QA snapshot reports **9,732 tests collected**, 0 pytest collection errors, and explicit documentation of accepted gates and unresolved historical blockers rather than presenting an artificially clean status.

Breakout Micro Following | MQL5 research system

github.com/bsAmirHeydar/breakout-micro-following

- Breakout and micro trend-following experiment with structured entries, regime filters, ATR-based risk logic, modular execution, and portfolio components.

Volatility Following | Volatility and fat-tail research toolkit

github.com/bsAmirHeydar/volatility-following

- Research toolkit for volatility regimes, fat tails, kurtosis, event structure, and volatility-aware systematic trading logic.

Desk | Fundamental research workspace

github.com/bsAmirHeydar/Desk

- Structured evidence-gathering and analytical workflow designed to make fundamental research repeatable across new markets and assets.
- Separates research question, evidence retrieval, method routing, inference, and final judgment; automation supports the process without receiving autonomous capital authority.

RESEARCH & TECHNICAL TOOLKIT

Markets	Gold (primary), Equities, FX, Commodities; strongest horizon: intraday/day trading
Research	Trend/momentum, mean reversion, breakout, volatility, regime-adaptive systems, discretionary-to-systematic translation
Quant / Risk	Expected value, hypothesis testing, fat tails, non-normal returns; CAGR, max drawdown, Calmar, profit factor, skew/tail, turnover, capacity, regime stability
Technology	Python for quantitative research and prototyping; Pandas, NumPy, SciPy, PostgreSQL, Docker; backtesting, data pipelines, APIs, testing, and research tooling; AI-augmented development workflow

DECISION & CAPITAL ORIENTATION

- Evaluates strategies first on mechanism, falsifiability, robustness, and failure modes; only then on portfolio fit, exposure, correlation, and capital scaling.
- Long-term direction is toward responsibility for strategy selection, risk ownership, and capital allocation, with quantitative and engineering systems supporting implementation and monitoring.

EDUCATION

High School Diploma

Independent study and project-based research in systematic trading, market structure, risk, probability/statistics, and algorithmic systems.